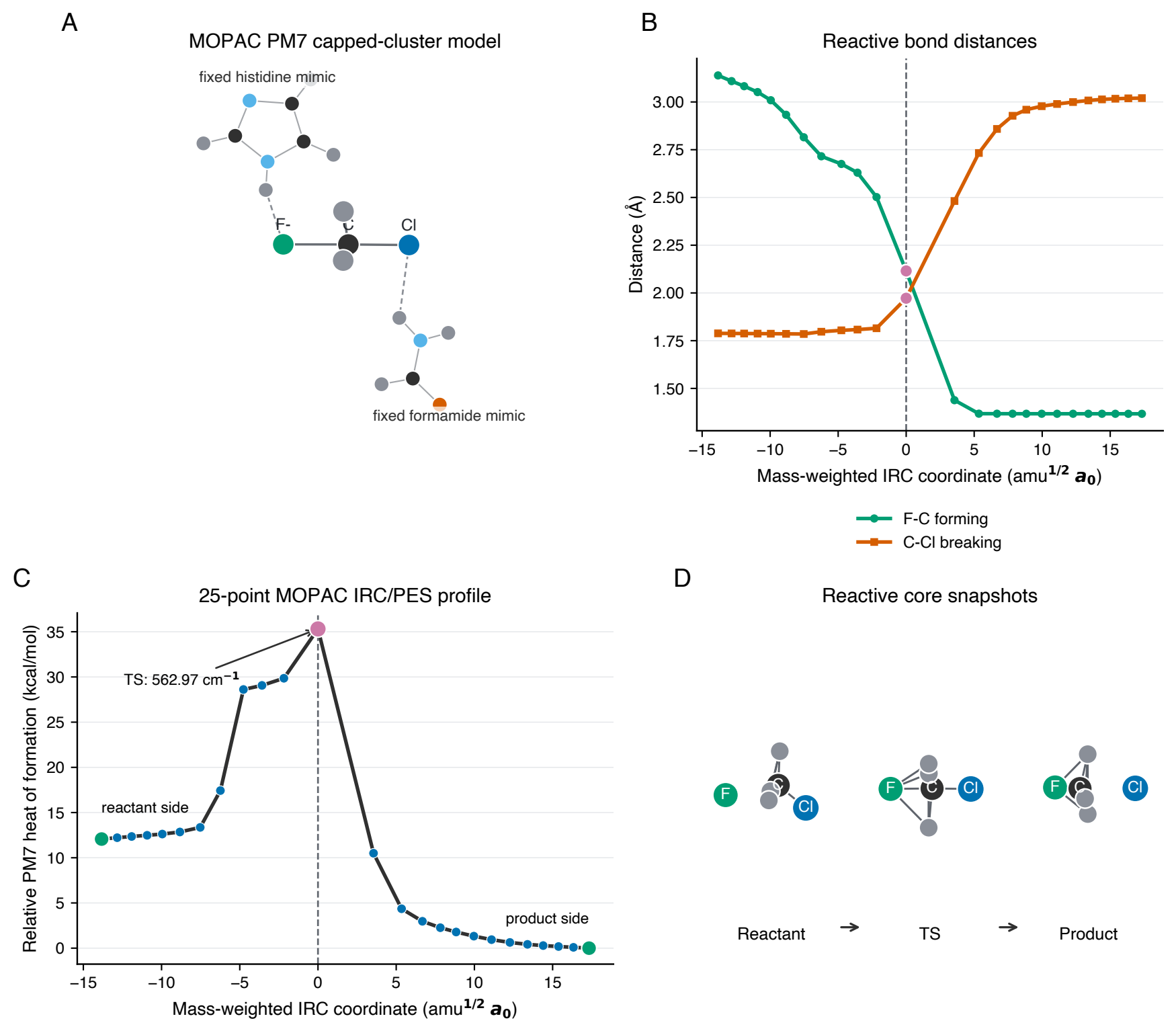
A
MOPAC PM7 capped-cluster model
fixed histidine mimic
F-
Cl
fixed formamide mimic
B
Reactive bond distances
Distance (Å)
3.00
2.75
2.50
2.25
2.00
1.75
1.50
−15
−10
−5
0
5
10
15
Mass-weighted IRC coordinate (amu$^{1/2}$ $a_0$)
F-C forming
C-Cl breaking
C
25-point MOPAC IRC/PES profile
Relative PM7 heat of formation (kcal/mol)
35
30
25
20
15
10
5
0
TS: 562.97 cm$^{-1}$
reactant side
product side
−15
−10
−5
0
5
10
15
Mass-weighted IRC coordinate (amu$^{1/2}$ $a_0$)
D
Reactive core snapshots
F
Cl
Reactant
TS
Product